Closed-Form Richardson-Number Stability Functions from Monin–Obukhov Similarity Theory: Re-Derivation, Generalization, and Implementation Notes

A Technical Working Reference for Boundary-Layer Modeling

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Abstract

This document provides a mathematically rigorous, fully self-consistent reference for transforming Monin–Obukhov Similarity Theory (MOST) flux-profile functions $\phi_m(\zeta)$ and $\phi_h(\zeta)$ into explicit, closed-form gradient Richardson number (Ri) stability functions $f_m(Ri)$ and $f_h(Ri)$. We document and correct a historical transcription error in the $\zeta \leftrightarrow Ri$ mapping relation found in early literature, highlight the general master equation governing the stable boundary layer (SBL) for arbitrary heat profile exponents α_h, detail the step-by-step derivation of the unstable branch cubic equation, derive near-neutral series expansions via series reversion, and provide an explicit integration of surface roughness z_0 for discrete boundary-layer models.

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1 Notation & Nomenclature

To reduce cognitive load and prevent ambiguity across local point-gradient theory, bulk atmospheric grid layers, and numerical implementations, the notation defined in Table 1 is maintained throughout this memorandum.

Table 1: Summary of Mathematical Symbols and Definitions

Symbol	Description	Definition / Relation
Ri_g	Gradient Richardson Number	$\frac{g}{\theta_0}\frac{\partial\theta/\partial z}{(\partial u/\partial z)^2} = \zeta\frac{\phi_h(\zeta)}{\phi_m^2(\zeta)}$
Ri_b	Bulk Layer Richardson Number	$\frac{g}{\theta_0}\frac{\Delta\theta\Delta z}{(\Delta u)^2}$
Ri_c	Critical Richardson Number	$\lim_{\zeta\to\infty} Ri_g(\zeta) = \frac{1}{\beta_m}$ (Limiting value for turbulence)
Ri	Generic Local Richardson Number	Used in general algebraic derivations ($Ri \equiv Ri_g$)
ζ	Dimensionless Monin–Obukhov Stability	z/L
L	Monin–Obukhov Length Scale	$\frac{u_*^2\theta_0}{\kappa g\theta_*}$
f_m, f_h	Ri-Based Stability Correction Functions	$K_m = l_m^2 S f_m(Ri), \quad K_h = l_h^2 S f_h(Ri)$
ϕ_m, ϕ_h	MOST Non-Dimensional Profile Functions	$\frac{\kappa z}{u_*}\frac{\partial u}{\partial z}, \quad \frac{\kappa z}{\theta_*}\frac{\partial\theta}{\partial z}$

2 Theoretical Foundations & Coupling Identicalities

In first-order eddy-diffusivity closures (K-theory), turbulent exchange coefficients for momentum (K_m) and heat (K_h) are parameterized using mixing-length theory driven by mean vertical shear $S = |\partial u/\partial z|$:

$$K_m = l_m^2 S f_m(Ri), \qquad K_h = l_h^2 S f_h(Ri) \tag{1}$$

where $l_m = l_h = \kappa z$ near the surface. Conversely, Monin–Obukhov Similarity Theory (MOST; Businger et al. 1971; Dyer 1974) defines diffusivities as:

$$K_m = \frac{\kappa z u_*}{\phi_m(\zeta)}, \qquad K_h = \frac{\kappa z u_*}{\phi_h(\zeta)} \tag{2}$$

Equating K-theory with MOST using the surface-layer profile relation $S = \frac{u_*}{\kappa z}\phi_m(\zeta)$ yields the fundamental coupling definitions:

$$f_m(Ri) = \frac{1}{\phi_m^2(\zeta)} \tag{3}$$

$$f_h(Ri) = \frac{1}{\phi_m(\zeta)\phi_h(\zeta)} = \frac{f_m^{1/2}(Ri)}{\phi_h(\zeta)} \tag{4}$$

2.1 $\zeta \leftrightarrow Ri$ Transformation

The gradient Richardson number Ri_g is defined via MOST profiles as:

$$Ri = \frac{g}{\theta_0}\frac{\frac{\partial\theta}{\partial z}}{\left(\frac{\partial u}{\partial z}\right)^2} = \zeta\frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} \tag{5}$$

Substituting (3) and (4) into (5) yields the exact relation connecting Ri and ζ:

$$\phi_m(\zeta) = f_m^{-1/2}(Ri), \qquad \phi_h(\zeta) = \frac{f_m^{1/2}(Ri)}{f_h(Ri)} \tag{6}$$

Inserting these back into (5):

$$Ri = \zeta\frac{f_m^{1/2}/f_h}{f_m^{-1}} = \zeta\frac{f_m^{3/2}(Ri)}{f_h(Ri)} \implies \boxed{\zeta = Ri\,\frac{f_h(Ri)}{f_m^{3/2}(Ri)}} \tag{7}$$

3 Stable Boundary Layer Formulation ($\zeta \geq 0$)

We generalize the stable non-dimensional profiles by retaining a linear momentum function ($\alpha_m = 1$) while allowing a general exponent α_h for heat:

$$\phi_m(\zeta) = 1 + \beta_m\zeta, \qquad \phi_h(\zeta) = (1 + \beta_h\zeta)^{\alpha_h} \qquad (\zeta \geq 0) \tag{8}$$

Defining $x \equiv f_m^{1/2}(Ri) = \phi_m^{-1}(\zeta)$, we invert $\phi_m(\zeta)$ to obtain:

$$\zeta = \frac{1-x}{\beta_m x} \tag{9}$$

3.1 Central Master Equation in $x \equiv f_m^{1/2}$

Substituting (9) into (7) gives the explicit expression for $f_h(Ri)$:

$$f_h(Ri) = \frac{x^2(1-x)}{\beta_m Ri} \tag{10}$$

Equating $\phi_h = x/f_h = \frac{\beta_m Ri}{x(1-x)}$ with $\phi_h(\zeta) = \left(1 + \beta_h \frac{1-x}{\beta_m x}\right)^{\alpha_h}$ produces the **central master equation** governing all stable FIRST-order closures:

$$\boxed{\left[\frac{Ri\,\beta_m}{x(1-x)}\right]^{1/\alpha_h} = \frac{\beta_m x + \beta_h(1-x)}{\beta_m x}} \tag{11}$$

3.2 Case 1: Standard Linear Heat Profile ($\alpha_h = 1$)

Setting $\alpha_h = 1$ reduces (11) directly to a quadratic polynomial in x:

$$Ri(\beta_h - \beta_m)x^2 - (1 + Ri\,\beta_h)x + 1 = 0 \tag{12}$$

Taking the physically meaningful root ($x \to 1$ as $Ri \to 0$) yields the exact closed-form momentum stability function:

$$\boxed{f_m(Ri) = x^2 = \left[\frac{(1 + Ri\,\beta_h) - \sqrt{(1 + Ri\,\beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)}\right]^2} \tag{13}$$

The corresponding heat function $f_h(Ri)$ is evaluated directly via (10).

3.3 Case 2: Businger–Dyer Quadratic Heat Profile ($\alpha_h = 2$)

Setting $\alpha_h = 2$ and squaring both sides of (11) clears fractional exponents, yielding a cubic polynomial $a_3x^3 + a_2x^2 + a_1x + a_0 = 0$:

$$\begin{aligned}
a_3 &= -(\beta_m - \beta_h)^2 \\
a_2 &= (\beta_m - \beta_h)^2 - 2\beta_h(\beta_m - \beta_h) \\
a_1 &= 2\beta_h(\beta_m - \beta_h) - \beta_h^2 - Ri\,\beta_m^3 \\
a_0 &= \beta_h^2
\end{aligned} \tag{14}$$

3.4 Equal-Parameter Limit ($\beta_m = \beta_h = \beta$)

When $\beta_m = \beta_h = \beta$, $a_3 = 0$ and $a_2 = 0$. The quadratic (12) degenerates to a simple linear equation in x:

$$(1 + \beta Ri)x = 1 \implies x = \frac{1}{1 + \beta Ri} \tag{15}$$

which yields:

$$\boxed{f_m(Ri) = (1 + \beta Ri)^{-2}, \qquad f_h(Ri) = (1 + \beta Ri)^{-3}} \tag{16}$$

Distinction Note: This result demonstrates that quadratic closures like $(1 - \beta Ri)^2$ stem from distinct physical assumption sets ($Pr_t = 1$ toy models) rather than standard linear MOST profile inversions.

4 Scalar Dissimilarity: Moisture versus Heat Functions

First-order closures often assume scalar similarity between sensible heat and water vapour. Observations show that this approximation can degrade under unstable, convective conditions, where coherent thermal plumes and moisture transport need not have identical profile functions. We therefore distinguish the moisture function ϕ_q from the heat function ϕ_h and define $r_{qT} \equiv \phi_q/\phi_h$. Values of r_{qT} different from unity should be treated as empirical, regime-dependent corrections rather than universal constants (Hill 1989; Mahrt 1991).

4.1 Moisture Stability Function

Adopting the same profile convention as for heat,

$$\phi_q(\zeta) = (1 - \beta_q \zeta)^{-\alpha_q}, \qquad f_q(Ri) = \frac{1}{\phi_m(\zeta)\phi_q(\zeta)} = \frac{f_m^{1/2}(Ri)}{\phi_q(\zeta(Ri))}. \tag{17}$$

Near neutrality and in stable conditions, $f_q \approx f_h$ is often an adequate approximation. Under strong convection, a separate calibration of (β_q, α_q) may be warranted; for example, $\beta_q \approx 12$ and $\beta_h \approx 16$ can be used as an illustrative sensitivity case, not as a universal parameter choice.

5 Unstable Boundary Layer Formulation ($\zeta < 0$)

For unstable stratification ($\zeta < 0$), standard Businger–Dyer profile functions take the generalized exponential form:

$$\phi_m(\zeta) = (1 - \beta_m \zeta)^{-\alpha_m}, \qquad \phi_h(\zeta) = (1 - \beta_h \zeta)^{-\alpha_h} \tag{18}$$

For classical coefficients ($\alpha_m = 1/4, \alpha_h = 1/2$), the stability functions map in terms of ζ as:

$$f_m(\zeta) = (1 - \beta_m \zeta)^{1/2} \tag{19}$$

$$f_h(\zeta) = (1 - \beta_h \zeta)^{1/2}(1 - \beta_m \zeta)^{1/4} \tag{20}$$

5.1 Master Cubic Equation in $x \equiv f_m^2$

Expressing f_h directly in terms of f_m:

$$f_h(f_m) = f_m^{1/2}\left[1 - \frac{\beta_h}{\beta_m}(1 - f_m^2)\right]^{1/2} \tag{21}$$

Substituting (21) into the corrected transformation identity $\zeta^2 = Ri^2 \frac{f_h^2}{f_m^3}$ and defining $x \equiv f_m^2(Ri)$ yields the exact master cubic equation for unstable conditions (see Appendix A for full intermediate steps):

$$\boxed{x^3 - 2x^2 + \left(1 - Ri^2\beta_m\beta_h\right)x - Ri^2\beta_m(\beta_m - \beta_h) = 0} \tag{22}$$

5.2 Analytical Solution for Equal Coefficients ($\beta_m = \beta_h = 16$)

When $\beta_m = \beta_h = 16$, the constant term in (22) vanishes. Factoring out $x \neq 0$:

$$x^2 - 2x + (1 - 256Ri^2) = 0 \implies (1 - x)^2 = 256Ri^2 \implies x = 1 - 16Ri \tag{23}$$

Recalling $x \equiv f_m^2$, we recover the analytical expression:

$$\boxed{f_m(Ri) = (1 - 16Ri)^{1/2}} \tag{24}$$

6 Series Reversions & Pedagogical Models

6.1 Near-Neutral Series Expansions via Series Reversion

When exact inversion is mathematically intractable, Taylor expanding $\phi_m(\zeta) = 1 + \beta_m\zeta + \gamma_m\zeta^2$ and $\phi_h(\zeta) = 1 + \beta_h\zeta + \gamma_h\zeta^2$ around $\zeta = 0$ provides a forward series for $Ri(\zeta)$:

$$Ri_g(\zeta) = \zeta + (\beta_h - 2\beta_m)\zeta^2 + (3\beta_m^2 - 2\beta_m\beta_h - 2\gamma_m + \gamma_h)\zeta^3 + O(\zeta^4) \tag{25}$$

Reverting this series yields the inverted expression for $\zeta(Ri_g)$:

$$\boxed{\zeta(Ri_g) = Ri_g + (2\beta_m - \beta_h)Ri_g^2 + (5\beta_m^2 - 6\beta_m\beta_h + 2\beta_h^2)Ri_g^3 + O(Ri_g^4)} \tag{26}$$

Substituting $\zeta(Ri_g)$ into $f_m = \phi_m^{-2}$ and $f_h = (\phi_m\phi_h)^{-1}$ yields the low-order Ri-series forms:

$$f_m(Ri_g) = 1 - \frac{1}{2}\beta_m Ri_g - \frac{1}{8}\beta_m(3\beta_m - 2\beta_h)Ri_g^2 + O(Ri_g^3) \tag{27}$$

$$f_h(Ri_g) = 1 - \frac{1}{2}(\beta_m + \beta_h)Ri_g + O(Ri_g^2) \tag{28}$$

6.2 One-Parameter Toy Model with Finite Ri_c

Under the explicit assumption of equal profile parameters ($\beta_m = \beta_h = \beta \implies Pr_t = 1$), the transformation reduces to $Ri_g = \frac{\zeta}{1+\beta\zeta}$. Inverting directly gives:

$$\zeta(Ri_g) = \frac{Ri_g}{1-\beta Ri_g} \implies \phi_m(Ri_g) = \frac{1}{1-\beta Ri_g} \tag{29}$$

6.3 Numerical Switch: Near-Neutral Taylor Expansion

For $|Ri|$ sufficiently small, evaluating a general cubic root can lose precision through cancellation as $x = f_m^2$ approaches unity. The series reversion above provides the local branch used by an operational implementation:

$$f_m(Ri) = 1 - \frac{1}{2}\beta_m Ri - \frac{1}{8}\beta_m(3\beta_m - 2\beta_h)Ri^2 + O(Ri^3). \tag{30}$$

For $\beta_m = \beta_h = 16$, this becomes

$$f_m(Ri) \approx 1 - 8Ri - 32Ri^2. \tag{31}$$

The threshold used in a numerical branch switch should be verified against the desired accuracy and floating-point implementation; $|Ri| \leq 0.01$ is a reasonable starting point, not a universal cutoff. Defining the critical Richardson number $Ri_c \equiv 1/\beta$, the closure functions reduce to:

$$\boxed{f_m(Ri_g) = f_h(Ri_g) = \left(1 - \frac{Ri_g}{Ri_c}\right)^2 = \left(\frac{Ri_c - Ri_g}{Ri_c}\right)^2} \tag{32}$$

7 Surface Layer Roughness (z_0) Integration

In numerical mesoscale and boundary-layer models (e.g., Louis 1979; McNider & Pielke 1981; England & McNider 1995), the finite vertical grid spacing requires calculating fluxes across a discrete layer from surface roughness height z_0 to first model level z_1. The bulk Richardson number across this layer is defined as:

$$Ri_b = \frac{g}{\theta_0} \frac{[\theta(z_1) - \theta(z_0)](z_1 - z_0)}{[u(z_1) - u(z_0)]^2} \tag{33}$$

Integrating continuous MOST profile equations $\frac{\partial u}{\partial z} = \frac{u_*}{\kappa z}\phi_m$ and $\frac{\partial \theta}{\partial z} = \frac{\theta_*}{\kappa z}\phi_h$ from z_0 to z_1 yields the integrated profiles:

$$u(z_1) = \frac{u_*}{\kappa}\left[\ln\left(\frac{z_1}{z_0}\right) - \Psi_m\left(\frac{z_1}{L}\right)\right], \qquad \theta(z_1) - \theta(z_0) = \frac{\theta_*}{\kappa}\left[\ln\left(\frac{z_1}{z_0}\right) - \Psi_h\left(\frac{z_1}{L}\right)\right] \tag{34}$$

Substituting these integrated profiles into (33) (assuming $z_1 \gg z_0$ such that $z_1 - z_0 \approx z_1$) expresses Ri_b in terms of integrated stability functions Ψ_m and Ψ_h:

$$Ri_b = \left(\frac{z_1}{L}\right) \frac{\ln(z_1/z_0) - \Psi_h(z_1/L)}{[\ln(z_1/z_0) - \Psi_m(z_1/L)]^2} \tag{35}$$

To ensure exact mathematical equivalence between the local point-gradient function $Ri_g = \zeta \frac{\phi_h}{\phi_m^2}$ and the discrete layer value Ri_b, an **effective depth** z_{eff} is defined such that $Ri_g(z_{\text{eff}}/L) = Ri_b$:

$$\boxed{z_{\text{eff}} = z_1 \left[\frac{\ln(z_1/z_0) - \Psi_h(z_1/L)}{[\ln(z_1/z_0) - \Psi_m(z_1/L)]^2} \right] \frac{\phi_m^2(z_{\text{eff}}/L)}{\phi_h(z_{\text{eff}}/L)}} \tag{36}$$

This relation is generally implicit because the stability functions depend on z_{eff}. In the neutral limit, it reduces to the explicit grid-matching height

$$z_{\text{eff, neutral}} = \frac{z_1}{\ln(z_1/z_0)}. \tag{37}$$

8 Dual Roughness Lengths ($z_{0,m}, z_{0,h}$) and Bulk Layer Integration

In atmospheric modeling, momentum transfer involves both skin friction and pressure drag over surface obstacles, whereas scalar transfer includes molecular and turbulent transport through the interfacial sublayer. Consequently, $z_{0,m} \neq z_{0,h}$ in general. The excess-resistance parameter $kB^{-1} \equiv \ln(z_{0,m}/z_{0,h})$ varies with surface type; values near 1–2 are common over relatively uniform surfaces, whereas substantially larger values can occur over rough, arid, or urban terrain (Garratt and Hicks 1973; Garratt 1992).

The integrated MOST profile functions from ground skin temperature $\theta_s = \theta(z_{0,h})$ to model level z_1 are:

$$u(z_1) = \frac{u_*}{\kappa} \left[\ln\left(\frac{z_1}{z_{0,m}}\right) - \Psi_m\left(\frac{z_1}{L}\right) + \Psi_m\left(\frac{z_{0,m}}{L}\right) \right] \tag{38}$$

$$\theta(z_1) - \theta_s = \frac{\theta_*}{\kappa} \left[\ln\left(\frac{z_1}{z_{0,h}}\right) - \Psi_h\left(\frac{z_1}{L}\right) + \Psi_h\left(\frac{z_{0,h}}{L}\right) \right] \tag{39}$$

When $z_1 \gg z_{0,m}, z_{0,h}$, the lower-limit corrections $\Psi_m(z_{0,m}/L)$ and $\Psi_h(z_{0,h}/L)$ are often neglected as a surface-layer approximation; retaining them is preferable when the roughness lengths are not small relative to $|L|$.

8.1 Bulk Richardson Number and Implicit Effective Depth

Incorporating dual roughness lengths into the discrete bulk Richardson number gives:

$$Ri_b = \frac{g}{\theta_0} \frac{[\theta(z_1) - \theta_s] z_1}{u(z_1)^2} = \left(\frac{z_1}{L}\right) \frac{\ln(z_1/z_{0,h}) - \Psi_h(z_1/L)}{\left[\ln(z_1/z_{0,m}) - \Psi_m(z_1/L)\right]^2} \tag{40}$$

Equating Ri_b to the local point-gradient function $Ri_g(z_{\text{eff}}/L) = \zeta \frac{\phi_h}{\phi_m^2}$ yields an implicit expression for the effective depth z_{eff} (Byun 1990):

$$z_{\text{eff}} = z_1 \left[\frac{\ln(z_1/z_{0,h}) - \Psi_h(z_1/L)}{\left[\ln(z_1/z_{0,m}) - \Psi_m(z_1/L)\right]^2} \right] \frac{\phi_m^2(z_{\text{eff}}/L)}{\phi_h(z_{\text{eff}}/L)} \tag{41}$$

Because the nondimensional stability functions ϕ_m and ϕ_h on the right-hand side depend on z_{eff}, (41) is generally solved by fixed-point iteration or a scalar root solver. Under neutral conditions ($\Psi_m = \Psi_h = 0, \phi_m = \phi_h = 1$), it reduces to the explicit form:

$$z_{\text{eff, neutral}} = z_1 \frac{\ln(z_1/z_{0,h})}{\left[\ln(z_1/z_{0,m})\right]^2} \tag{42}$$

9 Extension Beyond the Surface Layer: Blackadar Mixing Length

Monin–Obukhov Similarity Theory strictly applies within the surface layer (lowest 10% of the boundary layer), where the mixing length grows linearly ($\ell = \kappa z$). To prevent unrealistically large eddy diffusivities aloft, Blackadar (1962) introduced an asymptotic mixing length bound. To remain consistent with $z_{0,m} \neq z_{0,h}$, momentum and heat mixing lengths can be specified separately:

$$\ell_m(z) = \frac{\kappa z}{1 + \frac{\kappa z}{\lambda_{0,m}}}, \qquad \ell_h(z) = \frac{\kappa z}{1 + \frac{\kappa z}{\lambda_{0,h}}} \tag{43}$$

where $\lambda_{0,m}$ and $\lambda_{0,h}$ are asymptotic mixing-length limits for the outer boundary layer. Blackadar (1962) established the classic neutral empirical relation:

$$\lambda_{0,m} \approx 0.00027\,\frac{G}{f} \tag{44}$$

where G is the geostrophic wind speed and f is the Coriolis parameter (yielding typical values $\lambda_{0,m} \approx$ 30–100 m). Operational models frequently adopt the simplification $\ell_h(z) = \ell_m(z)$, though retaining a distinct $\lambda_{0,h}$ allows independent tuning of free-atmosphere heat exchange.

9.1 Coupling MOST Stability Functions with Bounded Mixing Lengths

Combining the bounded mixing lengths (43) with local stability functions $f_{m,h}(Ri)$ yields generalized eddy exchange coefficients for an outer-layer extension of the surface-layer closure. With $S \equiv |\partial u/\partial z|$:

$$K_m = \ell_m^2(z)\, S\, f_m(Ri) = \left(\frac{\kappa z}{1 + \frac{\kappa z}{\lambda_{0,m}}}\right)^2 \left|\frac{\partial u}{\partial z}\right| f_m(Ri) \tag{45}$$

$$K_h = \ell_m(z)\ell_h(z)\, S\, f_h(Ri) = \left(\frac{\kappa z}{1 + \frac{\kappa z}{\lambda_{0,m}}}\right)\left(\frac{\kappa z}{1 + \frac{\kappa z}{\lambda_{0,h}}}\right) \left|\frac{\partial u}{\partial z}\right| f_h(Ri) \tag{46}$$

Near the ground ($z \ll \lambda_{0,m}/\kappa, \lambda_{0,h}/\kappa$), $\ell_{m,h}(z) \to \kappa z$, recovering pure MOST. Aloft ($z \gg \lambda_{0,m}/\kappa, \lambda_{0,h}/\kappa$), $\ell_{m,h}(z) \to \lambda_{0,(m,h)}$, bounding eddy diffusivities independently of height.

10 Assumptions, Validity Domains, and Limitations of MOST

When implementing MOST-based closures in atmospheric models, students must recognize the foundational physical postulates and limitations of the theory:

1. **Quasi-Stationarity ($\partial/\partial t \approx 0$):** Turbulent flow statistics are assumed to be in local equilibrium. Rapid transients (e.g., gravity waves, frontal boundaries, rapid diurnal transitions) violate this balance.

2. **Horizontal Homogeneity ($\partial/\partial x = \partial/\partial y = 0$):** The surface is planar and uniform. Spatial advection and micro-scale land-use changes introduce flux divergences not captured by 1D MOST.

3. **Constant Flux Layer ($\partial\tau/\partial z \approx 0,\ \partial H/\partial z \approx 0$):** Vertical fluxes of momentum ($\tau = \rho u_*^2$) and heat ($H = \rho c_p u_* \theta_*$) vary by less than 10% across the surface layer.

4. **Negligible Coriolis Force ($z \ll u_*/f$):** Rotational effects are small compared to turbulent stress gradients within the surface layer ($z \lesssim$ 50–100 m), justifying the constant flux assumption.

5. **Universal Similarity Scaling (Monin & Obukhov 1954):** Surface-layer turbulence structure is uniquely governed by four dimensional quantities: friction velocity u_*, kinematic surface heat flux $Q_0 = u_*\theta_*$, buoyancy parameter g/θ_0, and height z. Shear production is not assumed to dominate a priori; mechanical and buoyant energy mechanisms are dynamically integrated via $\zeta = z/L$.

10.1 Key Physical Breakdown Regimes

- **Strongly Stable Boundary Layer ($\zeta \gg 1, Ri \to Ri_c$):** Under intense nocturnal cooling, buoyancy extinguishes continuous turbulence. Flow becomes intermittent and non-local, dominated by internal gravity waves and submeso drainage winds. Local gradient scaling (Nieuwstadt 1984) replaces surface MOST.

- **Unstable Free Convection Limit ($\zeta \to -\infty$):** As mean wind shear vanishes ($S \to 0, u_* \to 0$), $Ri \to -\infty$. Standard K-theory predicts $K_{m,h} \to 0$, whereas buoyant plumes drive strong vertical transport. The convective velocity scale $w_* = \left(\frac{g}{\theta_0} Q_0 z_i\right)^{1/3}$ replaces u_*, requiring non-local convective closures (e.g., Holtslag and Boville 1993).

- **The Roughness Sublayer ($z \leq 2$–$5h_c$):** Within vegetation or urban canopy environments of height h_c, individual wake eddies dominate. The flux-gradient relationship breaks down until reaching the inertial sublayer well above h_c.

11 Non-Stationarity and Nocturnal Low-Level Jets

MOST assumes quasi-steady, locally equilibrated turbulence. During nocturnal boundary-layer development, surface decoupling and inertial oscillation can produce a low-level jet (LLJ) above a shallow inversion (Blackadar 1957; Van de Wiel et al. 2010). The jet maximum can contain strong shear while the surface stress remains weak, so a local surface-layer closure should not be interpreted as a complete model of the jet or its turbulence.

In operational models, local stability functions can still provide the diffusive exchange coefficients within prognostic, time-dependent equations. For horizontally homogeneous flow, the coupled horizontal momentum equation may be written compactly as

$$\frac{\partial \boldsymbol{u}}{\partial t} + f\,\boldsymbol{k} \times (\boldsymbol{u} - \boldsymbol{u}_g) = \frac{\partial}{\partial z}\left(K_m(Ri)\frac{\partial \boldsymbol{u}}{\partial z}\right), \qquad \boldsymbol{u} = (u, v). \tag{47}$$

This coupling permits inertial rotation and LLJ formation while retaining MOST-based vertical diffusion where its local assumptions remain defensible.

12 Roughness Sublayer Extensions

Within the roughness sublayer, approximately $z_0 \le z \le z_{\mathrm{RSL}}$ with $z_{\mathrm{RSL}} \approx 2$–$5h_c$, wakes from canopy or urban elements modify the flux–gradient relationship. Standard MOST functions should therefore be regarded as an inertial-sublayer approximation in this region (Garratt 1980; Harman and Finnigan 2007).

An empirical RSL correction can be represented by

$$\phi^*_{m,h}(\zeta, z/h_c) = \phi_{m,h}(\zeta)\,\widehat{\phi}_{m,h}(z/h_c), \tag{48}$$

where one illustrative, smoothly matched ansatz is

$$\widehat{\phi}(z/h_c) = \exp\left[-c_1\left(1 - \frac{z}{z_{\mathrm{RSL}}}\right)^{c_2}\right], \qquad z < z_{\mathrm{RSL}}, \tag{49}$$

with c_1 and c_2 determined from observations or a canopy model. This ansatz is not universal; it is included to show how an RSL correction enters the closure. With $\widehat{\phi}_{m,h} = 1$ above z_{RSL}, the standard MOST functions are recovered continuously. The corresponding Richardson number and exchange functions are obtained from the modified profiles rather than by applying an unverified multiplicative correction to f_m and f_h.

A Complete Intermediate Derivation of the Unstable Cubic Equation

This appendix provides every intermediate algebraic step for deriving the unstable master cubic equation (22).

Step 1: Squaring the Corrected Transformation Identity. Starting from the corrected relation (7):

$$\zeta = Ri\,\frac{f_h(Ri)}{f_m^{3/2}(Ri)} \implies \zeta^2 = Ri^2\frac{f_h^2(Ri)}{f_m^3(Ri)} \tag{A1}$$

Step 2: Expressing ζ in terms of f_m. For Businger–Dyer momentum profile $\phi_m = (1-\beta_m\zeta)^{-1/4}$, equation (19) gives:

$$f_m = (1-\beta_m\zeta)^{1/2} \implies f_m^2 = 1-\beta_m\zeta \implies \zeta = \frac{1-f_m^2}{\beta_m} \tag{A2}$$

Step 3: Expressing f_h^2 in terms of f_m. For heat profile $\phi_h = (1-\beta_h\zeta)^{-1/2}$, equation (20) gives:

$$f_h = (1-\beta_h\zeta)^{1/2}(1-\beta_m\zeta)^{1/4} = (1-\beta_h\zeta)^{1/2} f_m^{1/2} \tag{A3}$$

Substituting $\zeta = \frac{1-f_m^2}{\beta_m}$ into $(1-\beta_h\zeta)$:

$$1-\beta_h\zeta = 1-\frac{\beta_h}{\beta_m}(1-f_m^2) = \frac{\beta_m-\beta_h(1-f_m^2)}{\beta_m} \tag{A4}$$

Squaring f_h:

$$f_h^2 = f_m\left[\frac{\beta_m-\beta_h+\beta_h f_m^2}{\beta_m}\right] \tag{A5}$$

Step 4: Substituting into Transformation Identity. Substituting ζ^2 from (A2) and f_h^2 from (A5) into (A1):

$$\left(\frac{1-f_m^2}{\beta_m}\right)^2 = Ri^2\frac{f_m\left(\frac{\beta_m-\beta_h+\beta_h f_m^2}{\beta_m}\right)}{f_m^3} \tag{A6}$$

Simplifying powers of f_m and β_m:

$$\frac{(1-f_m^2)^2}{\beta_m^2} = \frac{Ri^2}{\beta_m f_m^2}\left[(\beta_m-\beta_h)+\beta_h f_m^2\right] \tag{A7}$$

Step 5: Change of Variable $x \equiv f_m^2$. Setting $x = f_m^2$:

$$\frac{(1-x)^2}{\beta_m^2} = \frac{Ri^2}{\beta_m x}\left[(\beta_m-\beta_h)+\beta_h x\right] \tag{A8}$$

Multiplying both sides by $\beta_m^2 x$:

$$x(1-x)^2 = Ri^2\beta_m\left[(\beta_m-\beta_h)+\beta_h x\right] \tag{A9}$$

Expanding the left-hand side:

$$x(1-2x+x^2) = x^3-2x^2+x \tag{A10}$$

Expanding the right-hand side and grouping terms:

$$x^3-2x^2+x = Ri^2\beta_m(\beta_m-\beta_h)+Ri^2\beta_m\beta_h x \tag{A11}$$

Bringing all terms to the left-hand side yields equation (22):

$$x^3-2x^2+(1-Ri^2\beta_m\beta_h)x-Ri^2\beta_m(\beta_m-\beta_h) = 0 \tag{A12}$$

This completes the formal proof.

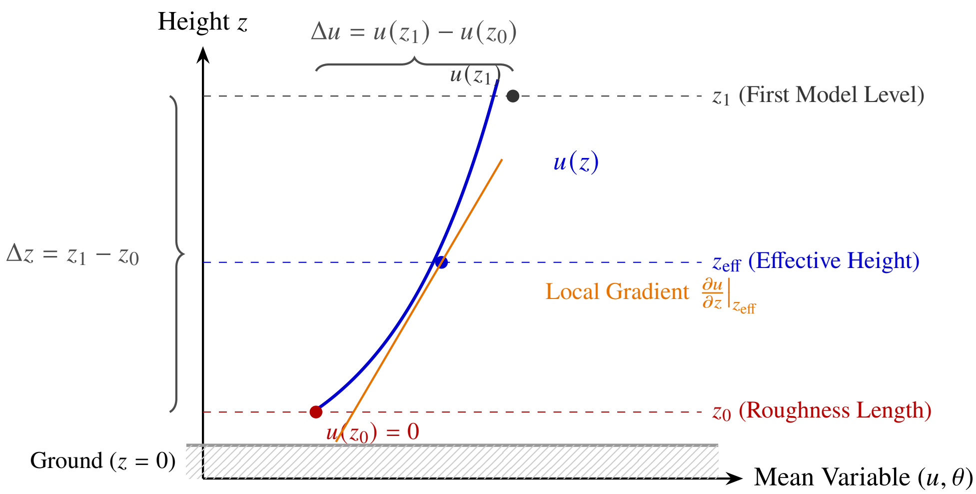

Figure 1: Schematic of discrete surface layer discretization illustrating the bulk layer differences over Δz and the effective height z_{eff} matching the local point gradient Ri_g to the bulk layer Ri_b.

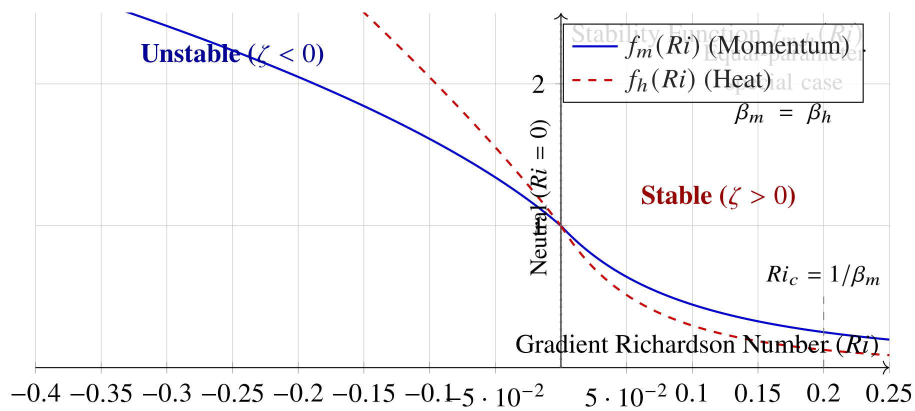

Figure 2: Illustration of the equal-parameter special case $\beta_m = \beta_h$ for the stable branch, where $f_m(Ri) = (1 + \beta Ri)^{-2}$ and $f_h(Ri) = (1 + \beta Ri)^{-3}$. This is not the general MOST stable solution; it is the reduced equal-parameter limit discussed in Section 8.

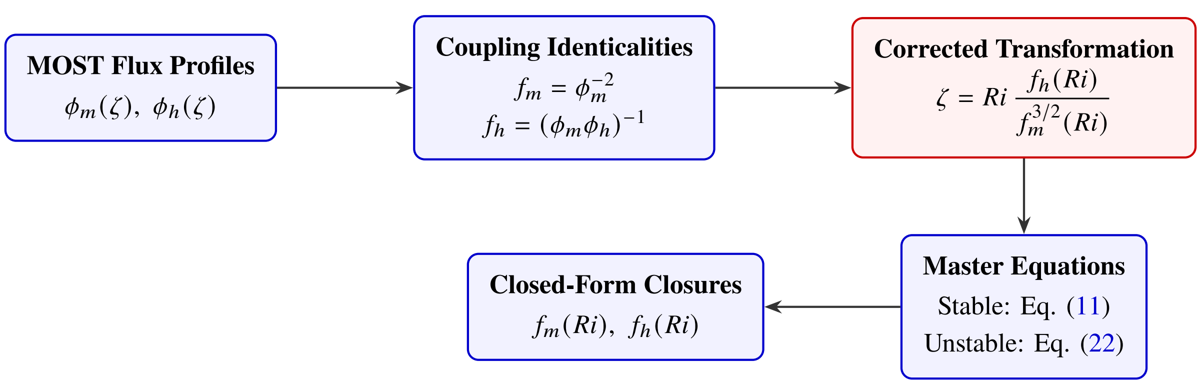

Figure 3: Mathematical workflow for transforming Monin–Obukhov Similarity Theory profile functions $\phi_{m,h}(\zeta)$ into analytical gradient Richardson number stability functions $f_{m,h}(Ri)$.

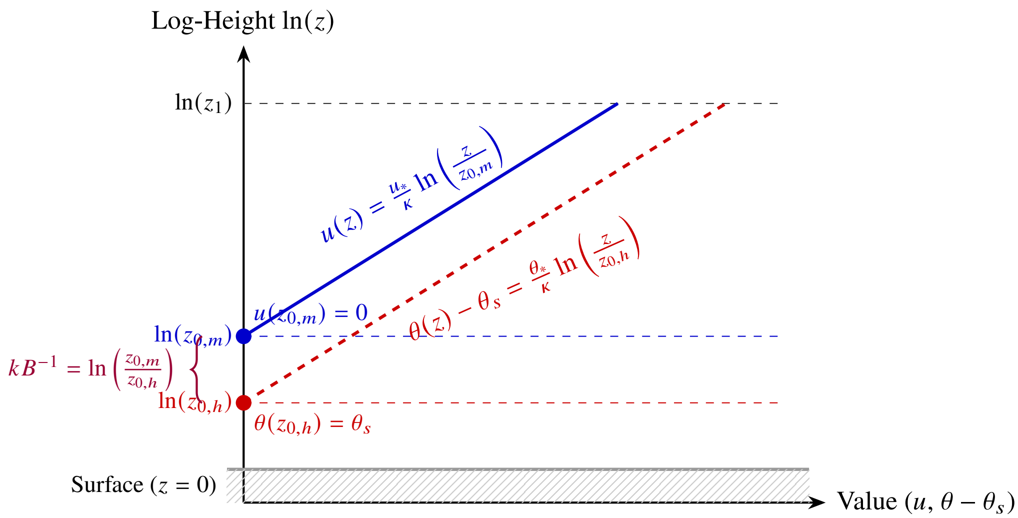

Figure 4: Neutral vertical profiles of wind velocity $u(z)$ and potential temperature deviation $\theta(z) - \theta_s$ plotted against logarithmic height $\ln(z)$. The logarithmic intercepts define $z_{0,m}$ (where wind vanishes) and $z_{0,h}$ (where atmospheric temperature matches ground skin temperature).

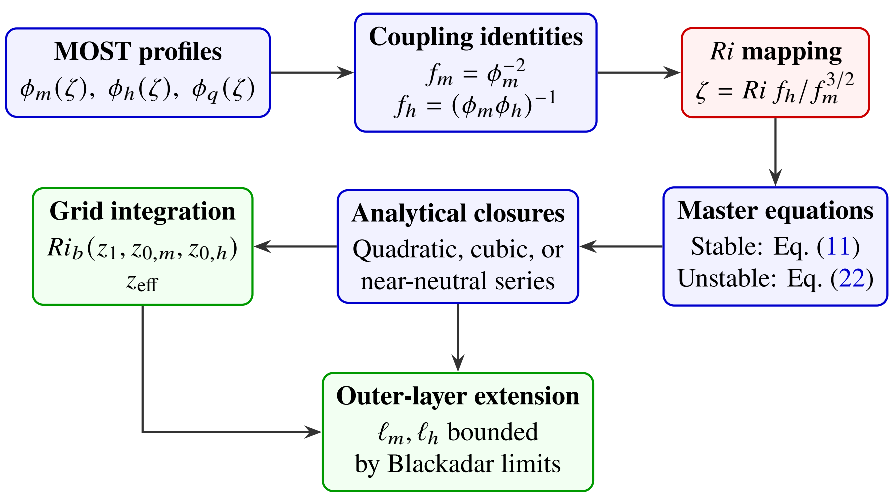

Figure 5: Expanded workflow from MOST profile functions through Richardson-number closures, dual-roughness grid integration, and bounded outer-layer mixing lengths. Scalar dissimilarity and non-stationary prognostic dynamics are additional modeling branches rather than consequences of local equilibrium MOST.

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